

A* Search

Time/Space Complexity: $O(b^d)$

Optimal: [2 conditions]

Admissible $\swarrow$ consistency $\searrow$

Complete: Yes
 it is guaranteed to find a solution path if one exists

$$f(n) = g(n) + h(n)$$

$\rightarrow$ g value
 $\rightarrow$ h value

$\rightarrow$ good value

Admissibility

$\rightarrow$ If the heuristic function is ~~admissible~~ admissible, then A* tree search will always find the least cost path $\Rightarrow$ least cost $\rightarrow$ $\text{नियम माला कोष्ट कोष्ट माला}$

$\rightarrow$ An admissible heuristic is one that never overestimates the cost to reach the goal.

$\rightarrow$ Admissible heuristics are by nature optimistic because, they think the cost of solving the problem is less than it actually is.

$\rightarrow$ St. line distance is admissible because, the shortest path betⁿ any ~~path~~ two points is a st. line.

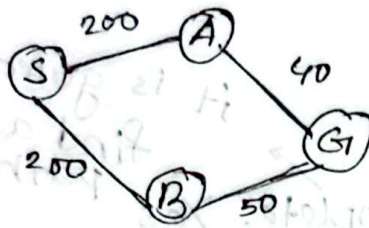
$\rightarrow$ Thus, St. line can't be overestimate

$$f(n) = g(n) + h(n)$$

$f(n)$ = The estimated total cost of the path through node(n)
 $g(n)$ = The actual cost or distance traveled from the start node to the current node.

$h(n)$ = The heuristic estimate the straight line distance from the current node n to the goal.

Underestimation:



$$h^*(A) = 40$$

$$h^*(B) = 50$$

$$f(A) = 200 + 30 = 230$$

$$f(B) = 200 + 20 = 220$$

$$h(A) = 30$$

$$h(B) = 20$$

G through B

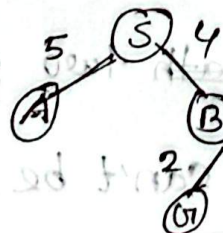
$$f(G) = 200 + 50 = 250$$

G through A

$$f(G) = 200 + 40 = 240$$

Underestimates always, provide optimal solution.

St. line	$h(n)$
S	2
A	3
B	2
G	0



- Path cost

Heuristic value

$$f(n) = g(n) + h(n)$$

$$\begin{aligned}
 f(A) &= g(A) + h(A) \\
 &= 5 + 3 \\
 &= 8
 \end{aligned}$$

▣ Node to Node minimum distance formula.

$$(y_2 - y_1)^2 + (x_2 - x_1)^2$$

↳ Uses straight-line distance

Importance

Searching $\rightarrow$ Lecture - 1 $\rightarrow$ Lecture $\rightarrow$ 2 . . .